

Reuel Joshua M

+91 8760378903 | reueljoshua2@gmail.com | [linkedin.com/in/reuel-joshua-m-48508a26ael](https://www.linkedin.com/in/reuel-joshua-m-48508a26ael)

SUMMARY

FPGA Engineer with 3+ years of experience in FPGA prototyping, hardware bring-up, system-level validation, and FPGA-based design analysis on Xilinx Ultrascale+ and Microchip Polar fire FPGA platforms. Strong understanding of digital design concepts including FSM design, FIFO architectures, Clock Domain Crossing (CDC) techniques, reset synchronization, and timing-aware design methodologies. Experienced in FPGA design flow including synthesis, implementation, timing analysis, resource optimization, design rule checks, and lint analysis. Hands-on experience with high-speed interface validation including Ethernet MAC, AXI interfaces, and GT transceiver integration. Skilled in FPGA debugging and validation using Vivado ILA, VIO, and IBERT, along with bare-metal C driver development for FPGA-based systems. Seeking FPGA Design and Validation opportunities to contribute to the development of high-performance and reliable FPGA solutions.

Technical Skills

- **FPGA Tools:** Xilinx Vivado, Vitis, Intel Quartus, Libero SoC.
- **Debugging Tools:** Vivado ILA, Vivado VIO, Intel Signal tap analyzer, Libero Identify Debugger.
- **Programming Languages:** Verilog HDL, C Programming.
- **Scripting Languages:** TCL Scripting (Basic)
- **Design Skills:** FPGA design, FSM designs, Clock Domain Crossing analysis, Reset Synchronization, Lint Analysis, Synthesis & Implementation Flow, Static Timing Analysis (STA), XDC Constraints analysis, Hardware Bring-up & Debugging.
- **Protocols & Interfaces:** Ethernet MAC_1G, Ethernet L2 Switch, UART, JTAG, RGMII, AXI, AHB, APB, IBERT and GT Transceiver interfaces.
- **FPGA Platforms:** Xilinx Kintex-7, Xilinx Ultrascale+, Microchip Polar fire FPGA platforms.
- **Operating Systems:** Linux, Windows.
- **Soft skills:** Problem Solving, Technical Documentation, Roots Cause Analysis, Team Collaboration, Cross-Functional Communication.

WORK EXPERIENCE

FPGA Engineer | SmartDV Technologies Private Limited

Jun 2023 – present

- Working on FPGA prototyping, hardware bring-up, and validation of FPGA-based systems using Xilinx Ultrascale and Microchip Polar Fire platforms.
- Involved in FPGA design analysis activities including RTL understanding, CDC analysis, Lint analysis, FIFO architectures, reset synchronization, and timing-aware design evaluation.
- Performed FPGA synthesis, implementation, timing analysis, XDC constraint management, and resource utilization analysis using Xilinx Vivado.
- Verified Ethernet MAC IPs, AXI/APB interfaces and GT transceiver-based high-speed communication.
- Debugged hardware issues using Vivado ILA, VIO, and IBERT tools.

PROJECTS

Ethernet MAC_1G | Tool: Xilinx Vivado

- Integrated and Validated Ethernet MAC_1G IP in Xilinx Kintex-7 Genesys2 FPGA platform using RGMII interface for Ethernet communication.
- Performed hardware bring-up and functional validation of Ethernet transmit and receive data paths, including packet transmission, reception, and frame integrity checks.
- Configured and validated MAC control/status registers through software-based register access and analyzed MAC functionality as per IP specification.
- Verified RGMII interface operation including TX/RX clocking, DDR data transfer, and MAC-PHY communication.

Ethernet L2 Switch | Tool: Xilinx Vivado & Libero SoC

- Validated Ethernet Layer-2 Switch IP functionality on FPGA platform including packet forwarding, MAC learning and traffic handling.
- Executed test scenarios for Ethernet packet transmission, ARP, ICMP, and data integrity verifications.
- Debugged Ethernet communication issues using Wireshark packet analysis and FPGA internal debug techniques.
- Analyzed PHY interface communication including RGMII connectivity and clock/data alignment issues.
- Verified design behavior against IP specification and supported issue debugging with RTL/design teams.

EDUCATION

Bachelor of Engineering - ECE
Anna University Regional Campus, Madurai

(CGPA: 8.2/10.0)